Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 39: Approved Solutions

Walks, Paths, and Connectivity

Covers Chartrand & Zhang, Section 12.2

Status of this set

Ungraded practice. Solutions released with it. This is the last practice set of the term.

Part I: How to Practice This Material

The four words, by what may repeat

Term	Vertices repeat?	Edges repeat?
walk	yes	yes
trail	yes	**no**
path	**no**	no
cycle	only start = end	no

A path has no repeated vertex, so it automatically has no repeated edge.

Three mistakes to expect

1. **Calling a trail a path.** Check vertices, not just edges.
2. **Forgetting isolated vertices when counting components.** Each is its own component.
3. **Assuming $n-1$ edges makes a graph connected.** Necessary, not sufficient.

Part II: Practice Problems

Section A — Vocabulary

1. In the graph with $V = \{v_1, \ldots, v_5\}$ and $E = \{v_1v_2, v_2v_3, v_3v_4, v_4v_1, v_2v_5\}$, classify each sequence as a walk only, a trail, a path, or a cycle.

(a) (v_1, v_2, v_3, v_4)

(b) $(v_1, v_2, v_3, v_4, v_1)$

(c) $(v_5, v_2, v_1, v_4, v_3, v_2)$

(d) (v_1, v_2, v_1, v_4)

Approved solution

(a) A path. No vertex repeats.
(b) A cycle of length 4. Only the start and end coincide.
(c) A trail but not a path. No edge repeats, but v_2 appears twice.
(d) A walk only. The edge v_1v_2 is used twice.

(c) is the one students misclassify, because the sequence looks like it wanders without backtracking. Check the *vertices*.

Section B — Components

2. For each graph, give $k(G)$ and say whether it is connected.

(a) $V = \{1, \ldots, 6\}$, $E = \{12, 23, 31, 45\}$

(b) $V = \{1, \ldots, 5\}$, $E = \{12, 23, 34, 45\}$

(c) $V = \{1, \ldots, 4\}$, $E = \{12, 34\}$

Approved solution

(a) $k(G) = 3$, disconnected. The components are $\{1, 2, 3\}$ (a triangle), $\{4, 5\}$, and $\{6\}$ — the isolated vertex 6 counts.
(b) $k(G) = 1$, connected. It is the path P_5.
(c) $k(G) = 2$, disconnected.

In (a), the vertex 6 appears in no edge and is easy to miss. Always count the vertices, not the edges.

3. A graph has order 9 and size 5.

(a) Can it be connected? Justify.

(b) What is the smallest number of components it could have?

Approved solution

(a) No. A connected graph of order n needs at least $n - 1$ edges, so order 9 requires at least 8. With only 5 edges it cannot be connected.
(b) Four. The bound $k(G) \geq n - m = 9 - 5 = 4$ gives a minimum of 4, and 4 is achievable — for example a path on 6 vertices (using 5 edges) together with 3 isolated vertices gives exactly 4 components.

Part (b) needs both halves: the bound shows 4 is the least possible, and a construction shows 4 actually occurs. A bound alone does not establish the minimum.

Section C — A proof

4. Prove that if a graph G has order n and every vertex has degree at least $\frac{n-1}{2}$, then G is connected.

Approved solution

Proof. Suppose, to the contrary, that G is disconnected. Then G has at least two components; let u and v lie in different components.
Every neighbour of u is in u's component, and every neighbour of v is in v's component, so these neighbour sets are disjoint and neither contains u or v themselves.
Counting vertices: u together with its at least $\frac{n-1}{2}$ neighbours accounts for at least $\frac{n+1}{2}$ vertices, and likewise for v. Since the two groups are disjoint, G has at least

$$\frac{n+1}{2} + \frac{n+1}{2} = n+1$$

vertices — contradicting that G has order n. ■

Proof by contradiction, with a counting argument inside it. The whole course in one proof: Lecture 27's method, Lecture 31's addition of disjoint counts, and this week's degree vocabulary.

Section D — Writing

5. In four to six complete sentences, explain why Theorem 12.26 — that a u–v walk guarantees a u–v path — is useful in practice, and describe the proof technique it uses.

Approved solution

Model answer. The definition of connectedness demands a *path* between every two vertices, and paths are constrained objects since no vertex may repeat. Walks are far easier to produce, because you may wander and backtrack freely. Theorem 12.26 says the easier object suffices: once you have exhibited any walk between two vertices, a path between them is guaranteed to exist, so a proof of connectedness never has to construct paths carefully. The proof takes a walk of smallest length and argues that if a vertex repeated, the portion between the repeats could be deleted to give a shorter walk — contradicting minimality. That is proof by contradiction applied to a minimal object, which works because the lengths are positive integers and so a smallest one exists.

Full credit requires both the practical point and the minimal-counterexample technique. The appeal to well-ordering is a bonus.

Instructor notes

- **Problem 4 is the best item in the set** — a contradiction proof with a counting argument inside. It shows students that the term's techniques compose.
- **Problem 3(b)** requires both a bound and a construction. Students routinely give only the bound.
- **Problems 1(c) and 2(a)** catch the two standard slips: trail-versus-path, and forgetting isolated vertices.
- **Problem 5** connects Theorem 12.26 to well-ordering from Lecture 30.
- **This is the final practice set.** Reserved for Homework 14: Section 12.2 even-numbered exercises.